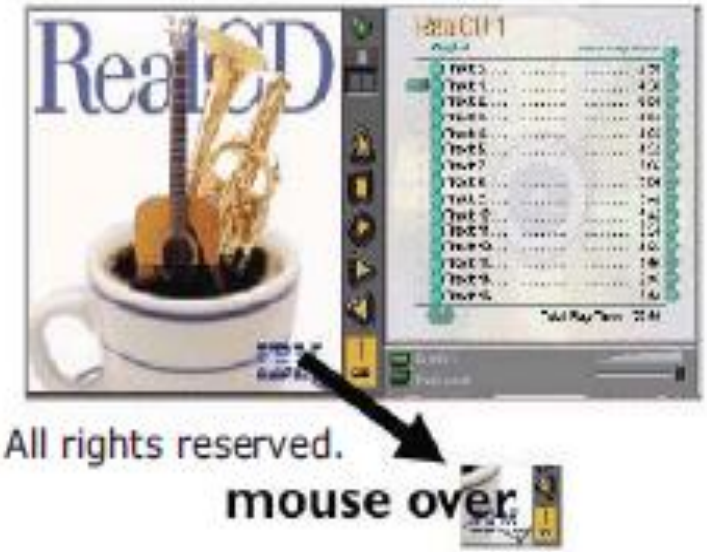
Visibility

Relevant parts of system should be visible

- Not usually a problem in the real world
- But takes extra effort in computer interfaces



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moma:workspheres - Mozilla

highlight

virtual
nomadic
domestic

hand-held home
stations
desktops
furniture
concepts

mini-exhibitions
objects
essays

photo essay

Natal Crescent

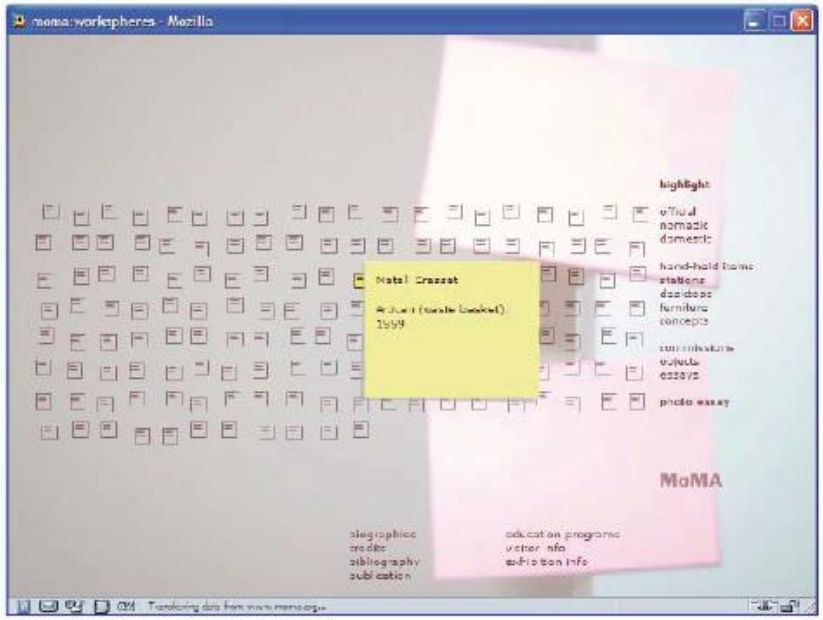
Architect (waste basket)
1969

MoMA

biographies
credits
bibliography
publication

education programs
visitor info
exhibition info

Translating data from www.moma.org...



metaphor: the interface represents a wall covered with Post-it notes, and you can zoom in on any one of them.

incredible simplicity. Nothing unnecessary was included. Note the use of whitespace to group the list of categories on the right, and the simple heading *highlight* that gives a clue to the function of the list

Unfortunately, finding something requires a lot of mouse waving. **«Mystery navigation»**

It's like old card game Concentration, demanding too much **recall** from the user, rather than offering easy **recognition** what you're looking for.

if real Post-it notes were arranged on a wall like this, you'd probably have just as much trouble navigating it. So the choice of metaphor may be the essence of the problem

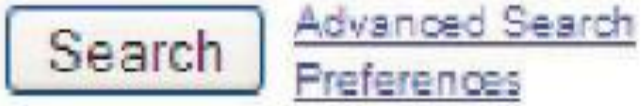
More “Mystery Navigation”

Today's Topics

- Visible actions
  - Information scent
- Visible state
  - Locus of attention
- Visible feedback
  - Perceptual fusion
  - Response time

Visibility of Actions: Use Appropriate Affordances

- Buttons & links



- Drop-down arrows

- Texture

- Mouse cursor

- Highlight on mouseover

Make Commands Visible

- Menus

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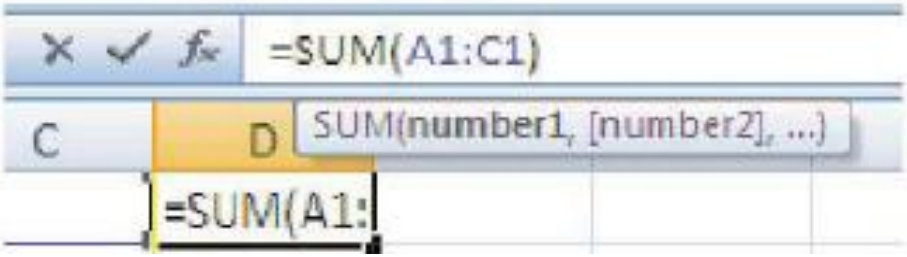
- Tooltips
- Self-disclosure



Information Scent

- Information foraging theory
  - Humans gathering information can be modeled like animals gathering food
  - Constantly evaluating and making decisions to maximize information collected against cost of obtaining it
- Information scent
  - Cues on a link that indicate how profitable it will be to follow the link to its destination

A link should smell like the content it leads to

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Localization ?

Bad scent include misleading terms (like “Set Program Access and Defaults” on the Windows XP Start menu)
Too-general labels (“Tools”)
Overlapping categories (“Customize” and “Options”)

Make Modes Clearly Visible

- Modes: states in which actions have different meanings
  - Vi's insert mode vs. command mode
  - Caps Lock
  - Drawing palette

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Visibility Depends on Locus of Attention

- **Spotlight metaphor**: attention focuses on one input channel (e.g. area of visual field) at a time

- Does the user's locus of attention include:
  - Caps Lock light on keyboard?
  - Status bar?
  - Menu bar?
  - Mouse cursor?

a user study mainly involving ordinary spreadsheet editing tasks, in which every five minutes the status bar would display "There's a $50 bill taped under your chair. Take it!"
In a full day of testing, more than a dozen users, nobody took the money
Phishing web sites exploit this to hide their stinky links

Visible Navigation State

- Breadcrumbs

Travel > Guides > North America

- Pagination

Results Page:
1 2 3 4 5 6 7 8 9 10 Next

- Tabs

Visible Model State

- Continuous visual representation of model
  - What to visualize should be guided by the user's tasks

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Visible View State

- Selection highlight

- Selection handles

- Drag & drop

Feedback:
Actions Should Have Immediately Visible Effects

- Low-level feedback
  - e.g. push button

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- High-level feedback
  - model state changes
  - new web page starts loading

Visibility vs. Security

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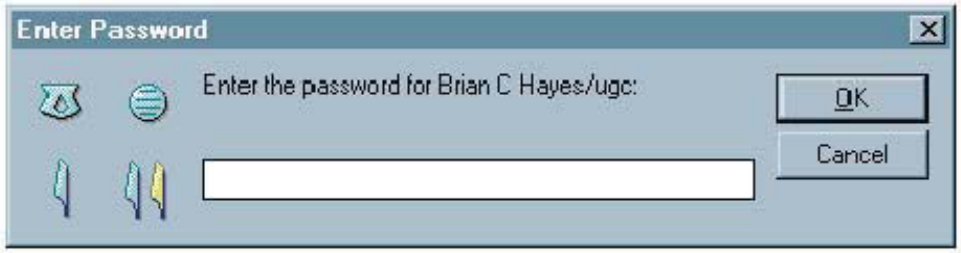

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a set of hieroglyphics derived from the password you typed in. If you can remember what hieroglyphics you usually see, then you can check them to see if you typed your password correctly

Perceptual Fusion

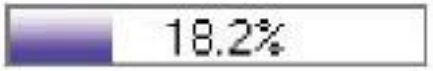
- Two stimuli within the same perceptual cycle (T_p ~ 100ms [50-200 ms]) appear **fused**
- Consequences
  - 1/ T_p frames/sec is enough to perceive a moving picture (10 fps OK, 20 fps smooth)
  - Computer response < T_p feels instantaneous
  - Causality is strongly influenced by fusion

Response Time

< 0.1 s: seems instantaneous

0.1-1 s: user notices the delay

1-5 s: display busy indicator

> 1-5 s: display progress bar 

Unnecessary Feedback

Visibility Isn't Just Seeing

- Audio
- Haptic

Case Study: Unix Command Line

- Invisible actions
- Invisible state
- No feedback on success

Windows cmd

```
C:\Intel\Logs>copy IntelAMT.log IntelAMT_.log
        1 file(s) copied.

C:\Intel\Logs>del IntelAMT_.log

C:\Intel\Logs>_
C:\Intel\Logs>del shjtgs
Could Not Find C:\Intel\Logs\shjtgs

C:\Intel\Logs>_
```

Visibility vs. Other Usability Principles

- Visibility primarily conveys information
- Supports learnability
- May conflict with simplicity

If you show too much – say, every property of an object, regardless of how useful it is – then the sheer volume of information may overwhelm, making the important bits **actually less visible** because they're surrounded so much

Summary

- Visibility of actions, state, and feedback
- Information scent
- Locus of attention affects visibility
- Perceptual fusion dictates response times